Erdos Fall 2026 Deep Learning Bootcamp



Box Office Smash or Bomb

Forecasting Movie Revenue with Machine Learning and Text Embeddings

Anthony Della Pella, Jacob Kapita, Marshall Scott, Pengfei Zhang

Introduction

Can we predict a film's box office before its ‘release?

$40M

Expensive bets

Median film budget, in 2025 dollars.

43%

Few films break even

of films with a known budget earned 2.5× it, the rough break-even point.

54%

Hits dominate

of all revenue goes to the top 10% of films.

Decisions come first. Greenlight, financing, marketing and release date are all decided before a single ticket is sold.

slidesmania.com

Goal

Predict a film's **worldwide revenue** from what is known before release:

Budget | Genre | Timing | Franchise | Studio | Cast & crew

…then turn it into a clear signal: **is this film worth producing?**

PROJECT STEPS

1. Combine & clean the data
2. Build pre-release features
3. Train models (LR, XGBoost)
4. Evaluate & explain features
5. **Profit & greenlight call**

slidesmania.com

Data Sources

Two Kaggle files describe the films; IMDb supplies the revenue label.

3,000 films

Kaggle TMDB · train

Budget, genres, cast, crew, keywords, release date; revenue as of 2019.

4,398 films

Kaggle TMDB · test

Same film details, but Kaggle never released their revenue.

63,249 titles

IMDb gross

Worldwide gross matched by IMDb id, so every film gets a revenue label.

7,398
films combined

→

−1,308
no revenue, pre-1980 or unreleased

→

6,090
labelled films, 1980–2017, in 2025 $

slidesmania.com

Data Overview

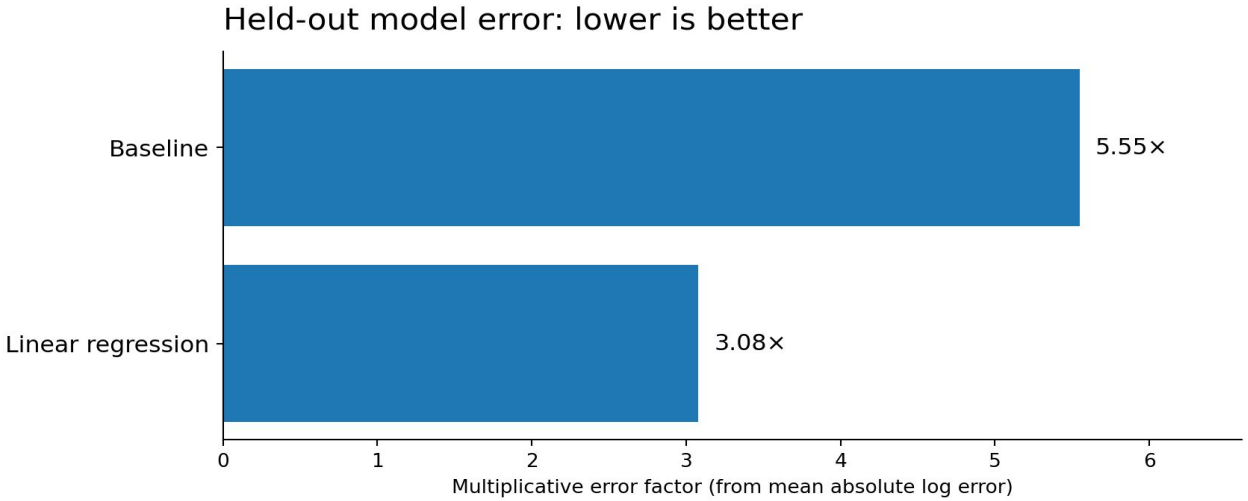
- IMDb-derived dataset: (1920–2026), used for genre comparisons, financial data cleaning, inflation and currency analysis, and a preliminary regression model.
- TMDB box-office dataset: training movies, with revenue labels, plus unlabeled competition test movies, used to explore budget/revenue patterns, movie clusters, production metadata, and imputation

slidesmania.com

EDA Finding 1

The first predictive benchmark is promising

A linear regression fit on 7,411 films and evaluated on 1,853 held-out films predicts the log of the domestic gross-to-budget ratio with test $R^2 = 0.498$. Its mean absolute error corresponds to approximately a 3.08x multiplicative factor, versus 5.55x for a constant-prediction baseline.



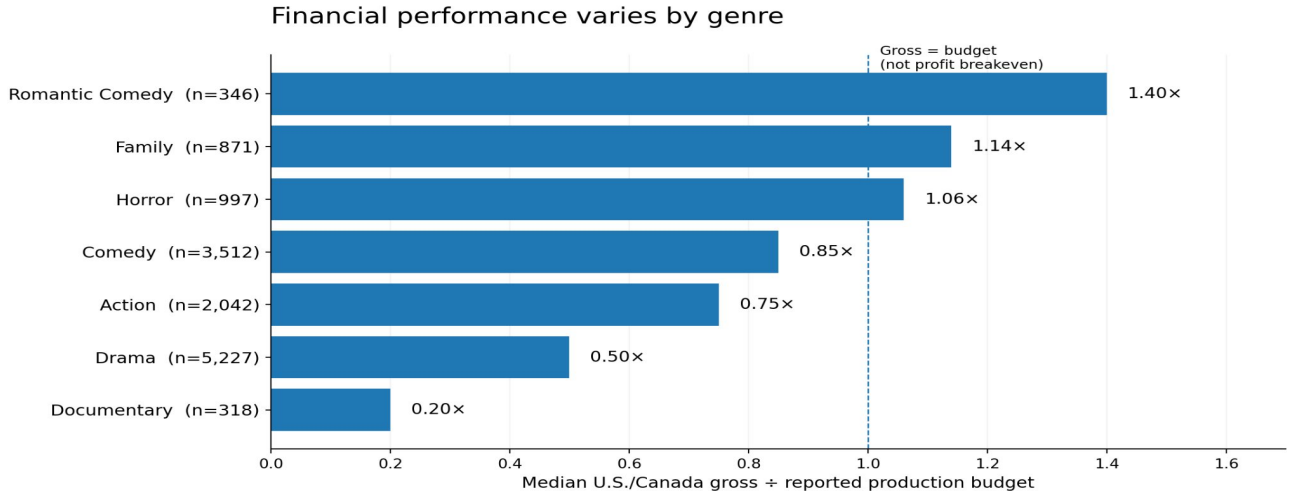
slidesmania.com

EDA Finding 2

The first predictive benchmark is promising

Domestic gross relative to budget varies by genre

Among the IMDb-derived films with valid N.A. box office and an identifiable budget of at least $10,000, the median N.A. gross-to-budget ratio is 0.67; 40.1% have a ratio greater than 1. Illustrative genre medians are romantic comedy (1.4), family (1.14), horror (1.06), drama (0.50), and documentary (0.20). *These are associations, not necessarily causal effects.*



slidesmania.com

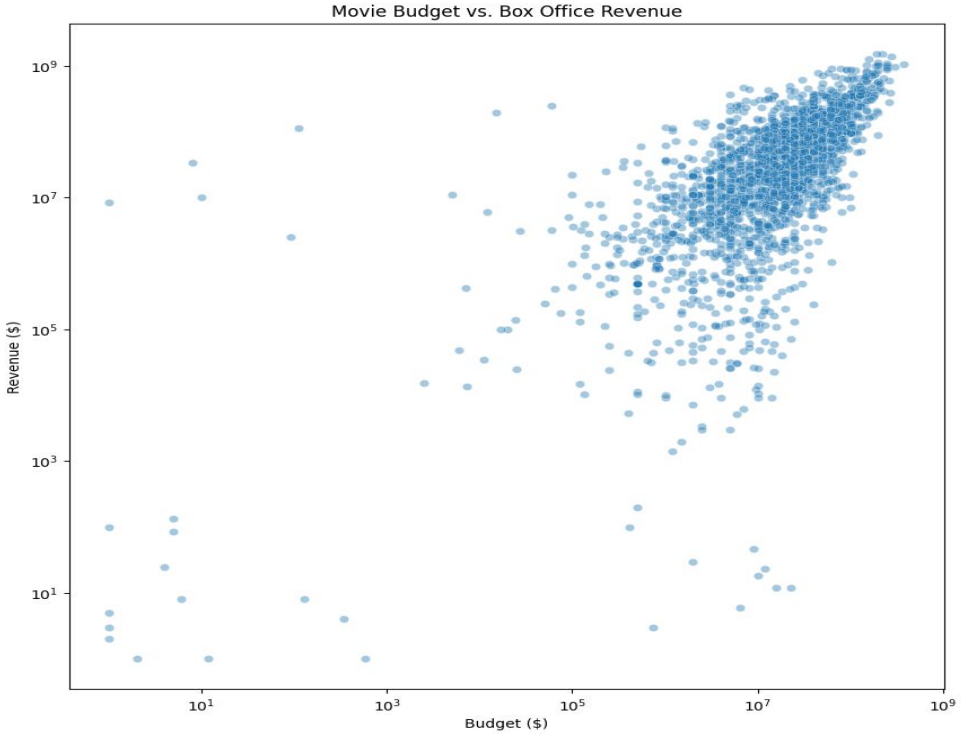
EDA Finding 3

The first predictive benchmark is promising

Domestic gross relative to budget varies by genre

Budget is a strong signal, not a guarantee

In the IMDb-derived dataset, production budget and worldwide box-office gross have a Spearman rank correlation of approximately 0.76 (computed for movies with both values). The separate TMDB sample shows a clear positive budget–revenue relationship, but also substantial scatter.



slidesmania.com

Modeling Approach

Goal: predict worldwide box office before release.

Data: 6,090 films (1980–2017): Kaggle TMDB data + IMDb worldwide grosses.

Target: ln(1 + revenue in 2025 dollars). The error metric is RMSLE.

Features: only information known before the release week.

- Structured: budget, runtime, release date, genre, language, country, sequel.
- Track record of cast and key crew, from earlier films only.
- Plot and keywords: use Sentence-BERT to turn them into numeric features.

Evaluation: 5-fold cross-validation on 5,176 films; final check on 914 test films.

slidesmania.com

Track Record Drives the Gain

5-fold cross-validation on 5,176 films · RMSLE, lower is better

Model	RMSLE (mean ± std)	vs linear baseline
Linear regression: baseline	1.553 ± 0.010	—
Linear regression: full	1.438 ± 0.008	+7.4%
XGBoost: baseline	1.435 ± 0.009	+7.6%
XGBoost: + track record	**1.372 ± 0.012**	**+11.7%**
XGBoost: + plot text	1.453 ± 0.015	+6.4%
XGBoost: + keywords	1.431 ± 0.018	+7.9%
XGBoost: full	**1.369 ± 0.015**	**+11.8%**

Track record lowers the error in all 5 folds; plot and keyword text add little.

slidesmania.com

Final Check on Held-out Films

914 test films, never used during development · one score per model

Model	Test RMSLE	vs linear baseline	Typical miss*
Linear regression: baseline	1.552	—	×3.2
XGBoost: baseline	1.406	+9.4%	×2.9
XGBoost: full	**1.350**	**+13.0%**	**×2.8**

- Test ≈ cross-validation (full model: 1.35 vs 1.37): no sign of overfitting
- A typical prediction is within a factor of about 3 of actual revenue
- Short of our 20% goal; budget and cast & crew track record do most of the work

slidesmania.com

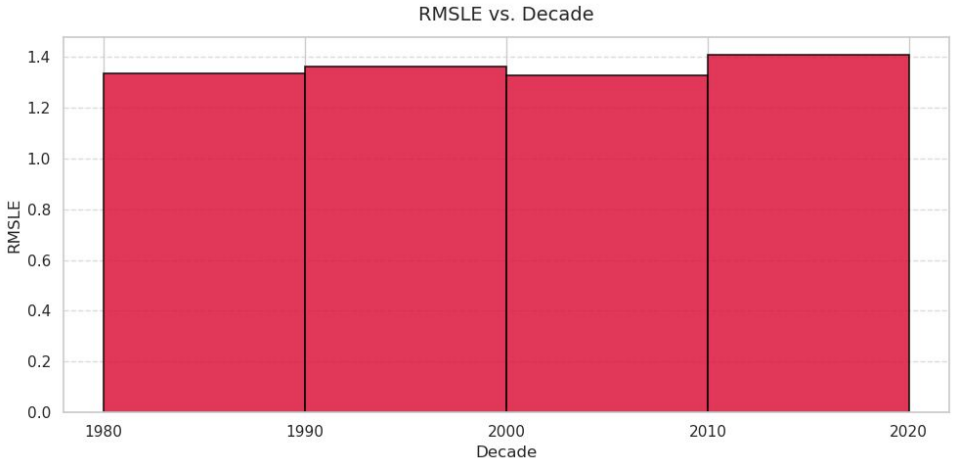
Model Performance

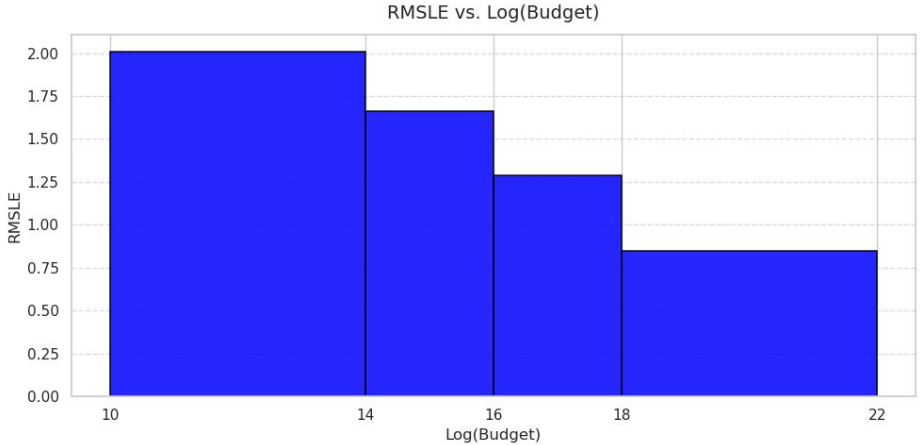
- Top model: XGBoost Full
- Model performance
  - RMSLE = 1.36
  - RMSLE with budget information = 1.24
  - Superior to the Linear regression
  - Consistent with being in top 10 TMDB Kaggle Box Office models
- Robust to missing data including budgets and unknown cast and crew members

slidesmania.com

Model Insights

- Accuracy
  - Constant over time
  - Increases as budget increases
- RMSLE near 1 for movie sequels

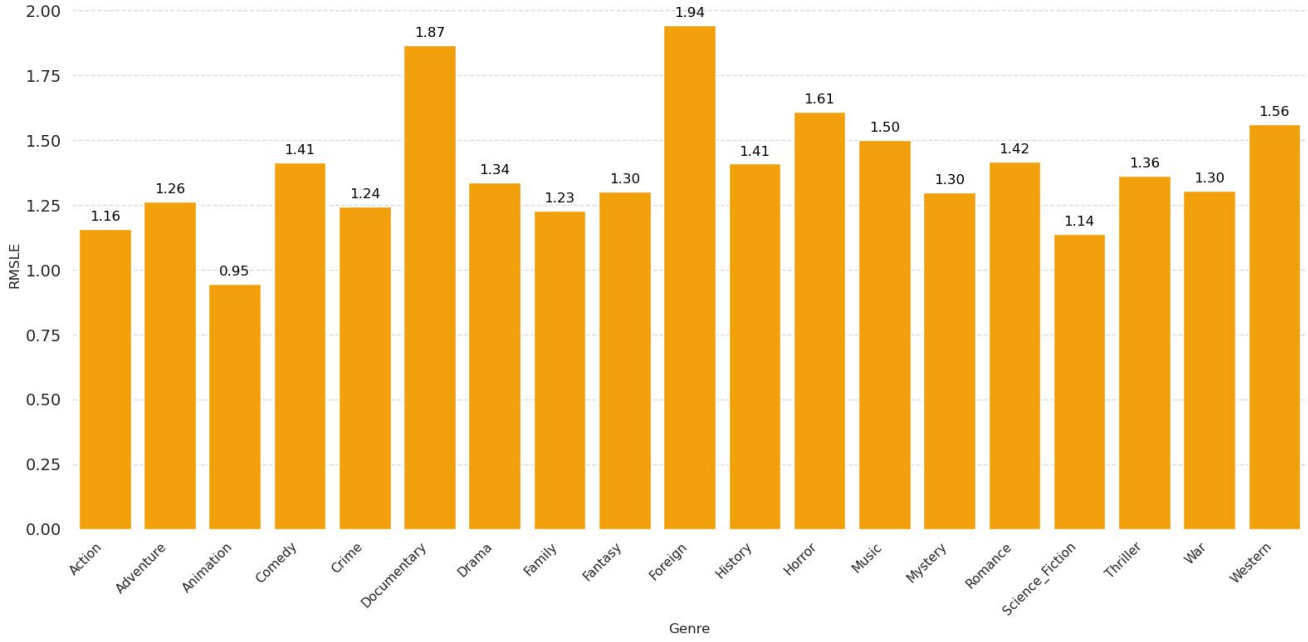




slidesmania.com

Model Insights

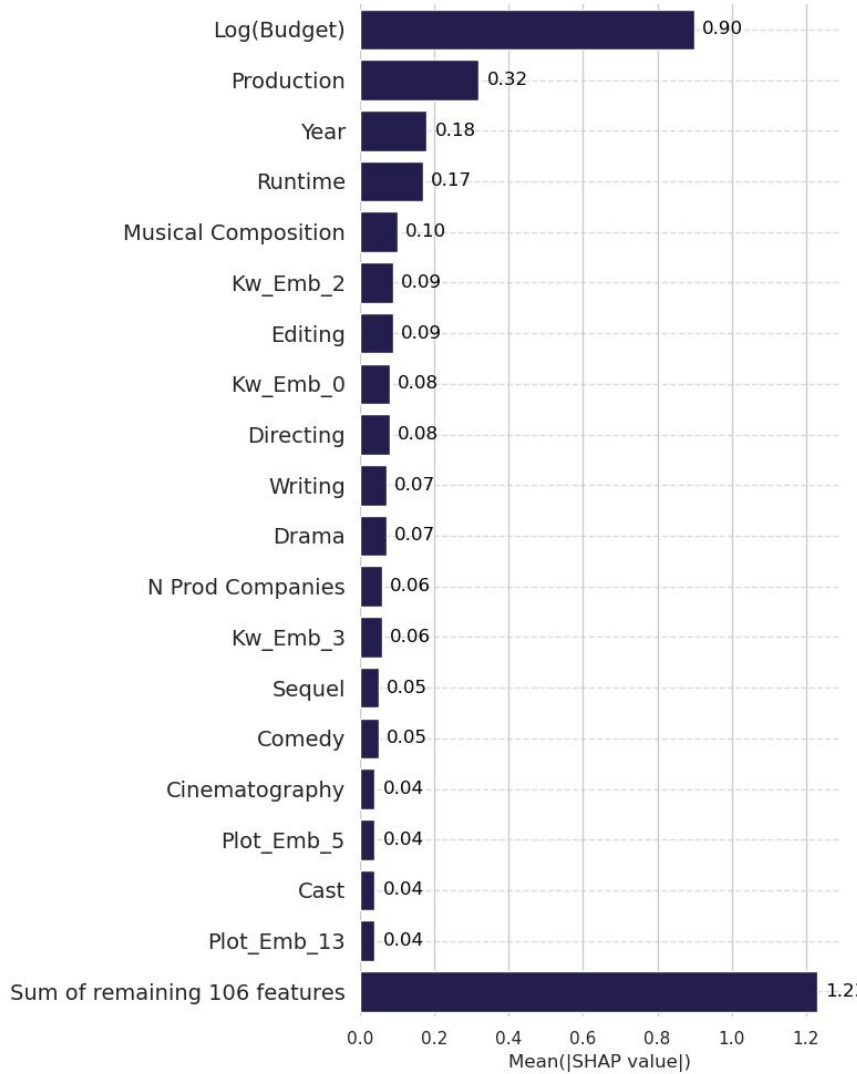
- Consistent across movie genres



slidesmania.com

Model Insights

- SHAP provided feature insights
  - Production, direction, editing, musical composition, writing, and cinematography were stronger predictors than cast.
  - Movies with genre tags for drama and comedy are also strong predictors.



slidesmania.com

Stakeholder Take away

- Our model can help production companies and producers make more profitable staffing decisions, mitigate financial risk, and improve investor confidence.
- Model improvement is possible
  - Expanding our embedding
  - Add post 2017 movie data
  - Include streaming and rental information

slidesmania.com